Sahaj Bhadja

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Summary

Frontend-focused Software Engineer with 2+ years building **React** and **TypeScript** single-page apps with real-time **WebSocket** updates on REST APIs, tested with Jest and Selenium, plus AWS, Linux, and Node.js and Python backend experience.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Cut **dashboard latency** from 1.2s to 350ms for 500+ users by restructuring the queries, indexes, and access patterns behind a multi-tenant platform's dashboard APIs on Azure SQL.
- Built FastAPI services and integrations with validation, error handling, and reconciliation checks, shipping CI/CD releases that kept 5K+ monthly jobs and 1K+ AI jobs at 99%+ production **reliability**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a **React** single-page application with upload, grading history, and question management views on Flask REST APIs, deployed to Vercel with GitHub Actions CI/CD for 150+ instructors and 10K+ daily requests.
- Designed the frontend-to-backend flow for file uploads, AI rubric generation, grading, and feedback on AWS RDS, processing 1K+ files per batch and cutting feedback turnaround 60% with stable **performance** at peak.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Wrote 20+ unit and integration **test** suites with PyTest and Mockito and automated nightly Jenkins runs on Linux, Windows, and macOS, using coverage reports and failure logs to debug cross-platform issues.
- Optimized **API** responses for low-bandwidth clients with pagination, page and limit parameters, and field filtering so clients fetched only needed data, improving response times 30% in Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **React** and TypeScript CRM interfaces with real-time **WebSocket** updates for resource allocation, billing, and leads on Node.js REST APIs, keeping 2M+ records synchronized across multiple locations.
- Automated Jest, Mocha, and Selenium unit, integration, and end-to-end **UI tests** in Jenkins CI/CD, reaching 95%+ build validation, and used Google Analytics load times and drop-off points to improve usability.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Designed end-to-end **UI/UX** flows in Figma and presented the architecture to the client, reducing design errors 25% and improving cross-platform compatibility 15% across iOS, Android, and web clients.
- Built booking APIs with Firebase **real-time** status sync and transactional locking, pushing live booking updates to React and React Native clients with sub-200ms latency and 40% faster confirmations.

Skills

Frontend: TypeScript, JavaScript, React, Next.js, HTML, CSS, TailwindCSS, Shadcn, Material UI, Angular
UI Engineering: Single-Page Applications, Reusable Components, Responsive Design, Real-Time UIs, Data Visualization, Performance Optimization
Testing & Debugging: Jest, Mocha, Selenium, PyTest, Unit Testing, Integration Testing, UI Testing, Code Reviews, Debugging
Backend & APIs: REST APIs, WebSockets, Node.js, Express.js, FastAPI, Flask, Python, Firebase
Cloud & Tools: AWS (EC2, S3, RDS), Linux, Git, GitHub Actions, CI/CD, Docker, Vercel, Sentry

Projects

FinNexus Live | *Next.js, TypeScript, WebSockets, TailwindCSS, MongoDB*

- Built a **TypeScript** and Next.js SPA with real-time WebSocket updates, Plaid bank linking, an advanced transaction filtering UI, and responsive TailwindCSS components, serving 50K+ users across 1M+ transactions.

Big-Brain Live | *React, Shadcn, Convex, Clerk, LLaMA-2*

- Built a collaborative document app with **real-time** multi-user sync via Convex, LLaMA-2 summarization and Q&A, and a responsive Shadcn UI, cutting document review time 60% for 200+ users.

HMIS – Hospital Management Information System Live | *TypeScript, React, TailwindCSS, Twilio, Sentry*

- Won Smart India Hackathon with a TypeScript and **React** app featuring doctor and hospital dashboards, appointment booking and rescheduling forms, and responsive TailwindCSS UI, monitored with Sentry across 100+ daily appointments.